

RStudio 和 R 語言 安裝步驟

安裝網址：<https://posit.co/download/rstudio-desktop/>

 PRODUCTS ▾ FREE & OPEN SOURCE ▾ USE CASES ▾ PARTNERS ▾ LEARN & SUPPORT ▾ ABOUT ▾

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 314.19 MB | [SHA-256: 84442594](#) | Version: 2026.01.1+403 | Released: 2026-02-19

R 語言安裝：(選擇自己的作業系統安裝)

The Comprehensive R Archive Network

Download and Install R

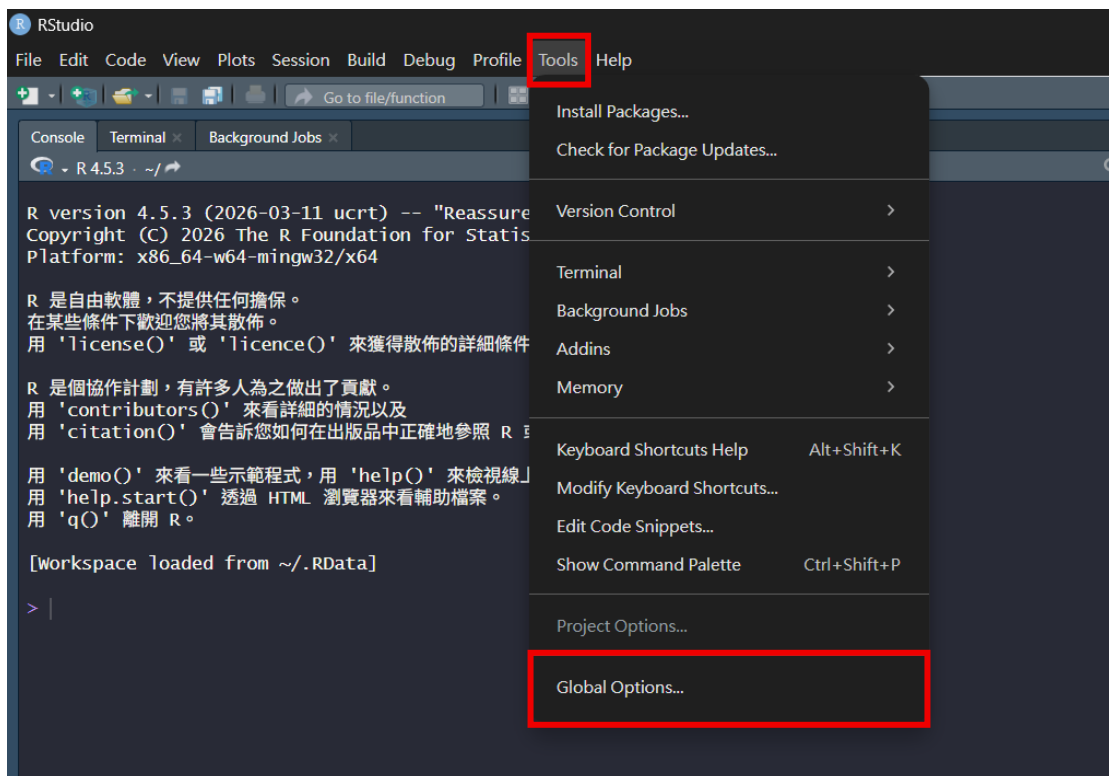
Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux \(Debian, Fedora/Redhat, Ubuntu\)](#)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

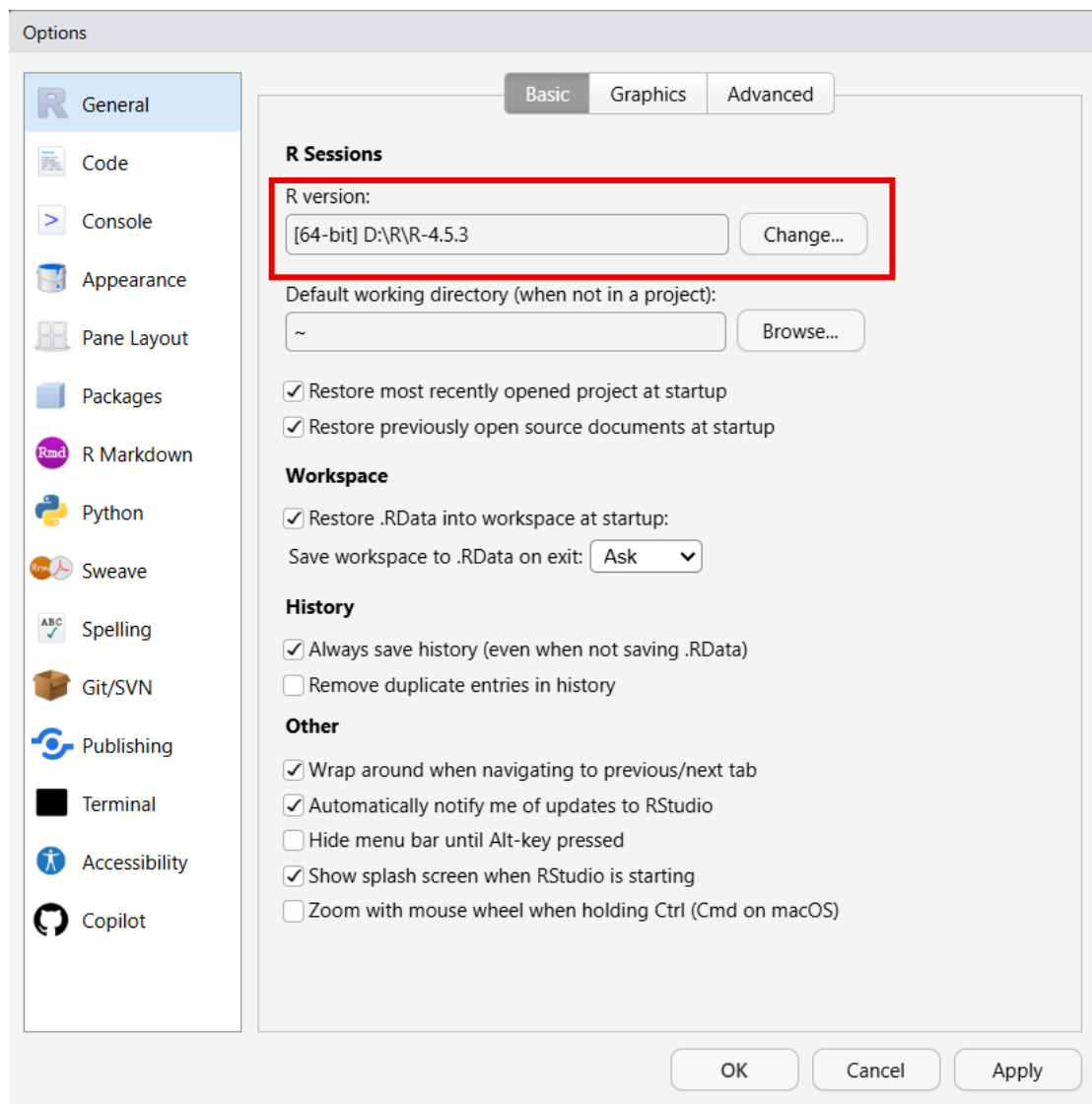
安裝完後，R studio 的設定(windows 系統)：

主要是選擇安裝的 R 版本



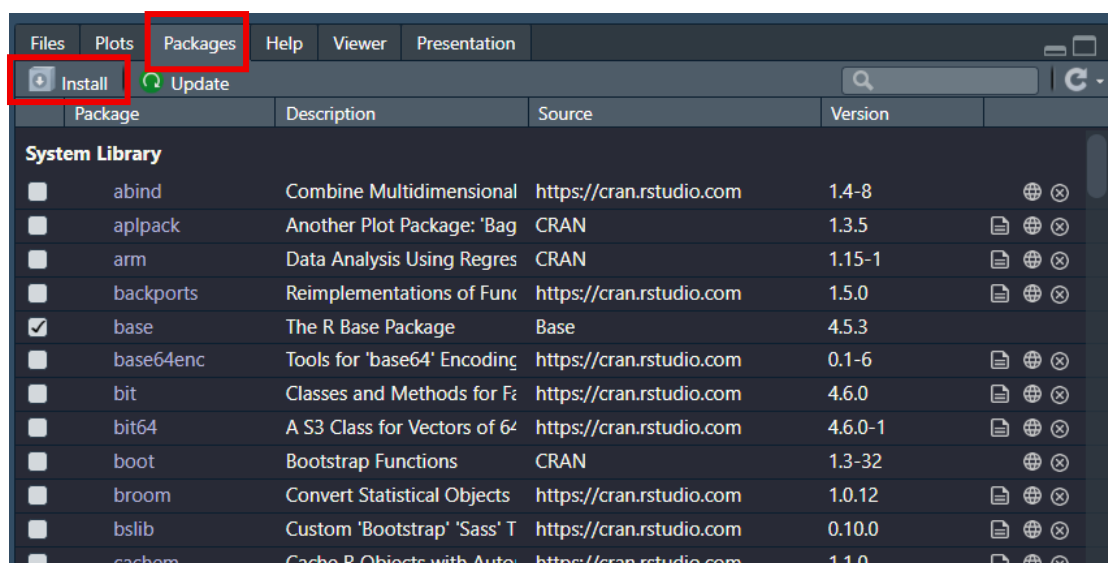
The screenshot shows the RStudio application window. The 'Tools' menu is open, and 'Global Options...' is highlighted at the bottom. The console shows the R version 4.5.3 (2026-03-11 ucrt) and the platform x86_64-w64-mingw32/x64. The console also displays the R license text in Chinese.

(選擇 R 語言安裝的地方)

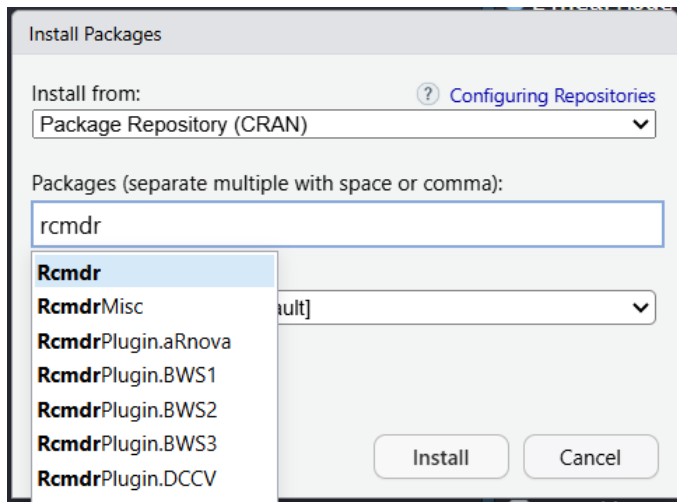


安裝 Rcmdr、RcmdrPlugin.KMggplot2、RcmdrPlugin.survival 套件

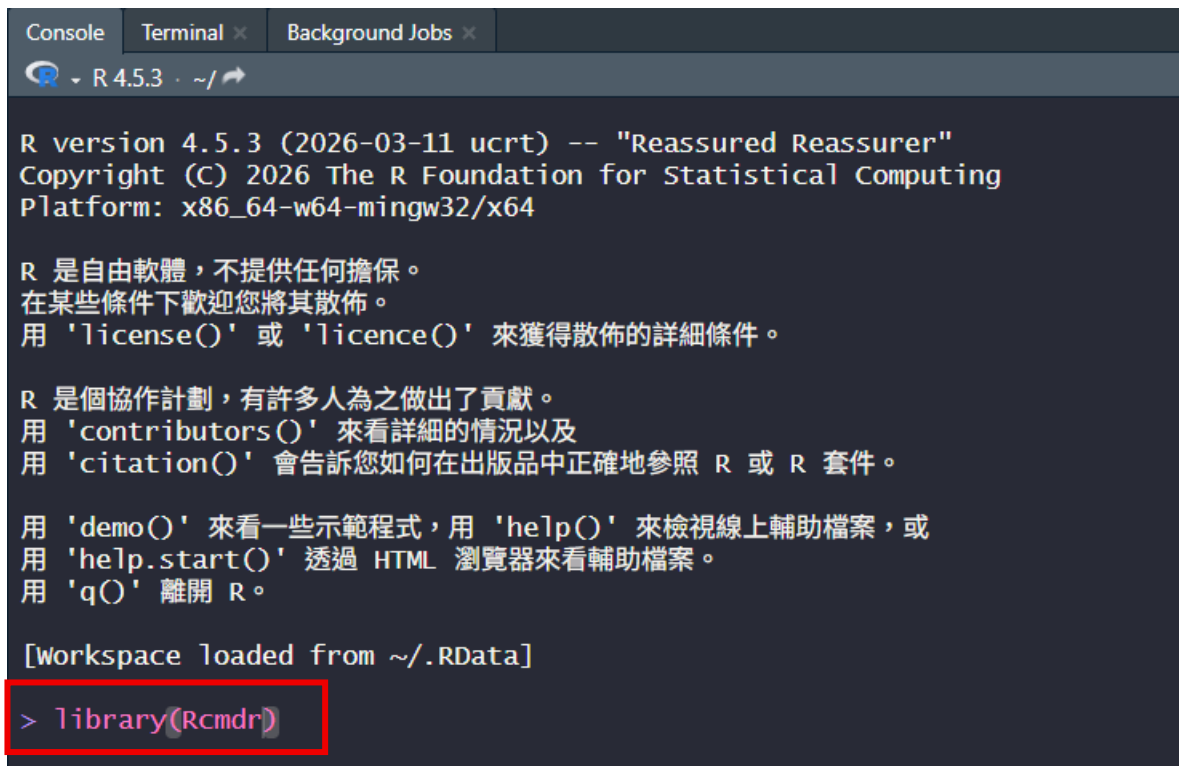
(為了要畫出生存曲線 survival curve 以及用 Coxregression model 分析 Hazard Ratio)



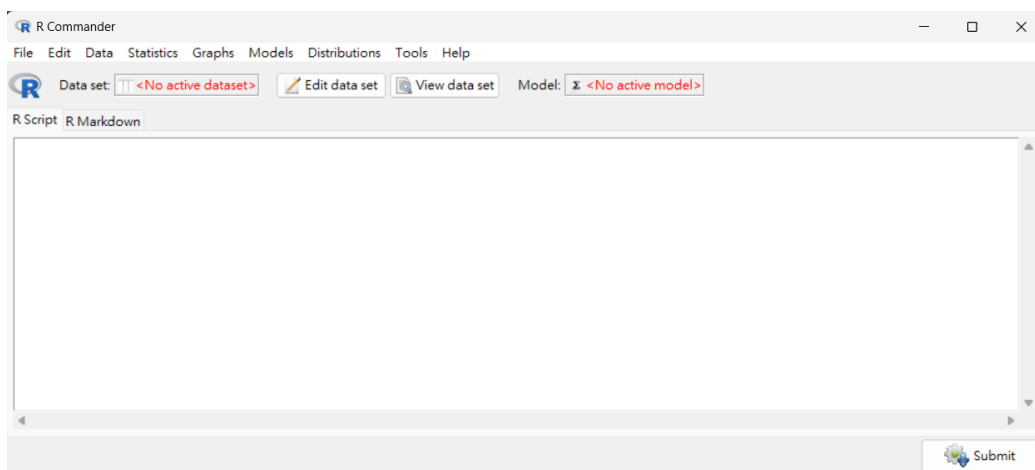
(依序下載套件)



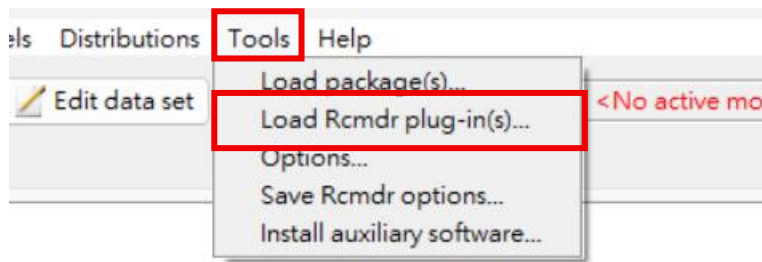
安裝完後叫出 Rcmdr



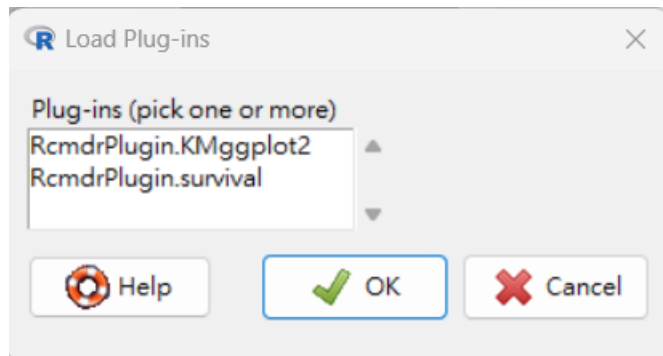
第一次叫出 Rcmdr 的時候會跳出小視窗，基本上就是按 OK 就對了
成功叫出的畫面



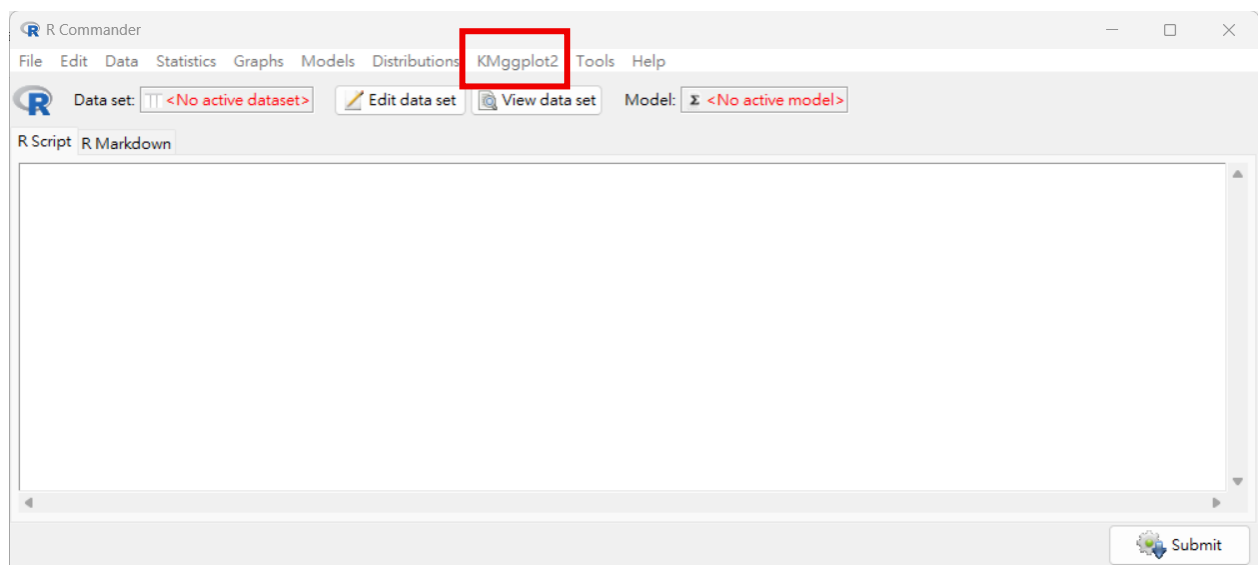
先載入安裝的 Plugin 套件(每次叫出 Rcmdr 都要再重新載入一次)



兩個都載入 (按 OK 之後，會需要重新啟動 Rcmdr 才能成功載入)

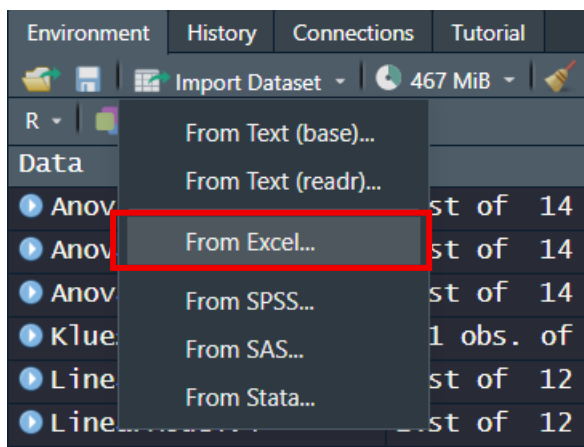
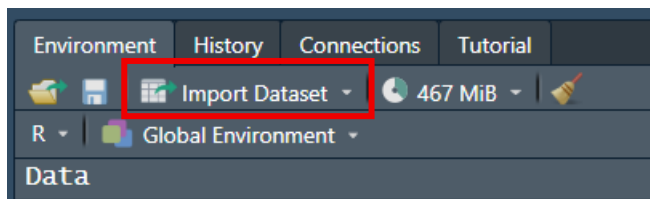


載入成功後會看到標籤列多了 KMggplot2

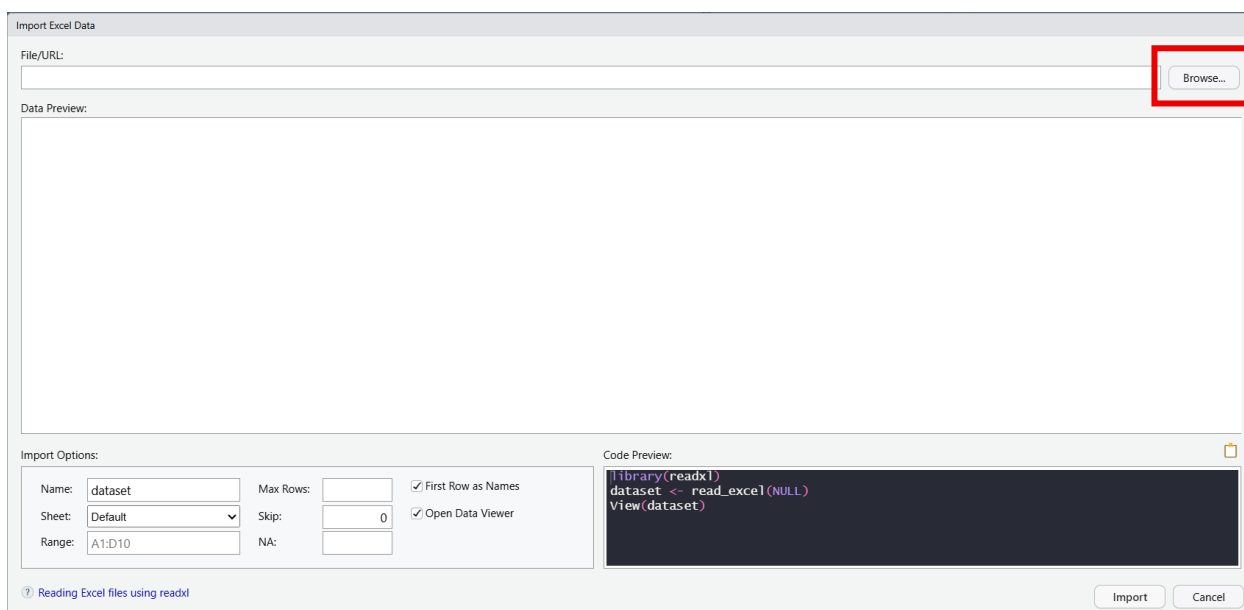


Survival Curve

最一開始從 RStudio 的右上方載入要分析的 excel 檔



選擇檔案，選擇完後點選 Import 載入檔案



Import Excel Data

File/URL:

/PDC_study_clinical_data_pdc000546_merge.xlsx

Browse...

Data Preview:

Case ID (character)	Case Submitter ID (character)	Genomic and Imaging Data Resource (character)	Ethnicity (character)	Sex (character)	Race (character)	Morphology (character)	Primary Diagnosis (character)	Site of Re...
0339e66-fc1a-448e-8786-5aeca5a2d4d5	C3L-07213	GDC: https://portal.gdc.cancer.gov/cases/53d5cace-2ade-48...	not hispanic or latino	female	white	9440/3	Glioblastoma	Tempor...
06187e41-2249-4c23-a284-1a97c9548bdc	C3N-03480	GDC: https://portal.gdc.cancer.gov/cases/2a73c63b-212b-4...	not reported	male	asian	9440/3	Glioblastoma	Tempor...
0a91df81-5338-4718-9578-e5e1a82bce1	C234192	GDC: https://portal.gdc.cancer.gov/cases/cabb8504-1154-4...	not reported	male	not reported	9440/3	Glioblastoma	Tempor...
0e776008-7475-4a1e-a188-be933245b0bd	C827913	GDC: https://portal.gdc.cancer.gov/cases/fe540737-d48f-49...	not reported	male	not reported	9440/3	Glioblastoma	Tempor...
20d515c6-c7cb-46b3-a5a8-a94759ee0bb3	C3N-03452	GDC: https://portal.gdc.cancer.gov/cases/f54ddb9-5251-4d...	not reported	male	white	9440/3	Glioblastoma	Frontal
219fe210-4592-4fd7-8e2c-45b2910ea300	C3N-04113	GDC: https://portal.gdc.cancer.gov/cases/a629e6ae-5baf-41...	not reported	male	asian	9440/3	Glioblastoma	Brain, h...
23a9255-d162-4274-8759-0b28b3267e1f	C3N-03755	GDC: https://portal.gdc.cancer.gov/cases/13226b4e-99e2-4...	not reported	male	white	9440/3	Glioblastoma	Tempor...
24ab2f1c-2630-4782-8d50-4dda20e02939	C3L-04083	GDC: https://portal.gdc.cancer.gov/cases/ca055248-952a-47...	not hispanic or latino	female	white	9440/3	Glioblastoma	Frontal
258df9a7-fbcb-4141-8584-d6d97ace6bf2	C3L-04038	GDC: https://portal.gdc.cancer.gov/cases/e2307982-2357-4...	not reported	female	white	9440/3	Glioblastoma	Parietal
26cd6af4-4aa2-4992-bf20-84418a330b18	C3L-02644	GDC: https://portal.gdc.cancer.gov/cases/ca540923-722b-4...	not reported	female	white	9440/3	Glioblastoma	Brain, h...
27c6f7bb-4ae0-4a76-8a9a-aecb4704b7aa	C3L-04359	GDC: https://portal.gdc.cancer.gov/cases/be171afc-eecl-43...	not reported	male	white	9440/3	Glioblastoma	Parietal

Previewing first 50 entries.

Import Options:

Name: PDC_study_clinical_data_px

Sheet: Default

Range: A1:D10

Max Rows:

Skip: 0

NA:

☒ First Row as Names

☒ Open Data Viewer

Code Preview:

```
library(readxl)
PDC_study_clinical_data_pdc000546_merge <- read_excel("F:/研究所/資料庫平台
/kegg_hmdb/PDC_clinicalData/PDC_study_clinical_data_pdc000546_merge.xlsx")
view(PDC_study_clinical_data_pdc000546_merge)
```

Import

Cancel

叫出 Rcmdr

Console Terminal Background Jobs

R 4.5.3 ~ /

R 是個協作計劃，有許多人為之做出了貢獻。
用 'contributors()' 來看詳細的情況以及
用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或
用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。
用 'q()' 離開 R。

[workspace loaded from ~/.RData]

> library(readxl)
> PDC_study_clinical_data_pdc000546_merge <- read_excel("F:/研究所/資料庫平台/PDC_clinicalData/PDC_study_clinical_data_pdc000546_merge.xlsx")
> view(PDC_study_clinical_data_pdc000546_merge)
> library(Rcmdr)

開啟 Rcmdr 後，確認已載入 KMGgplot2

R Commander

File Edit Data Statistics Graphs Models Distributions **KMGgplot2** Tools Help

Data set: <No active dataset>

Edit data set

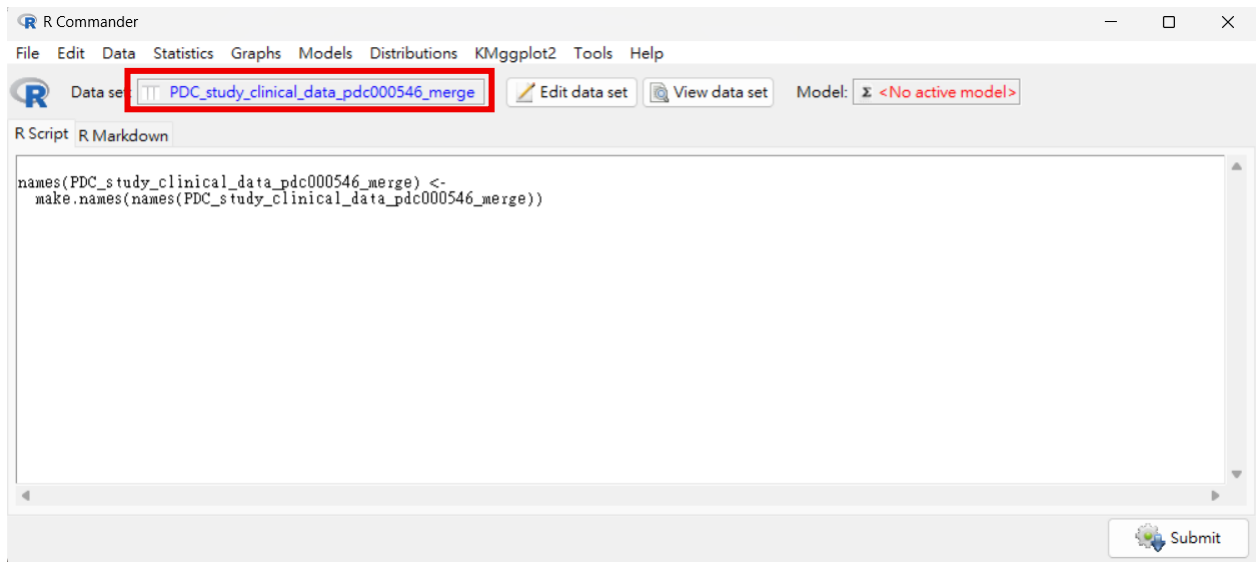
View data set

Model: <No active model>

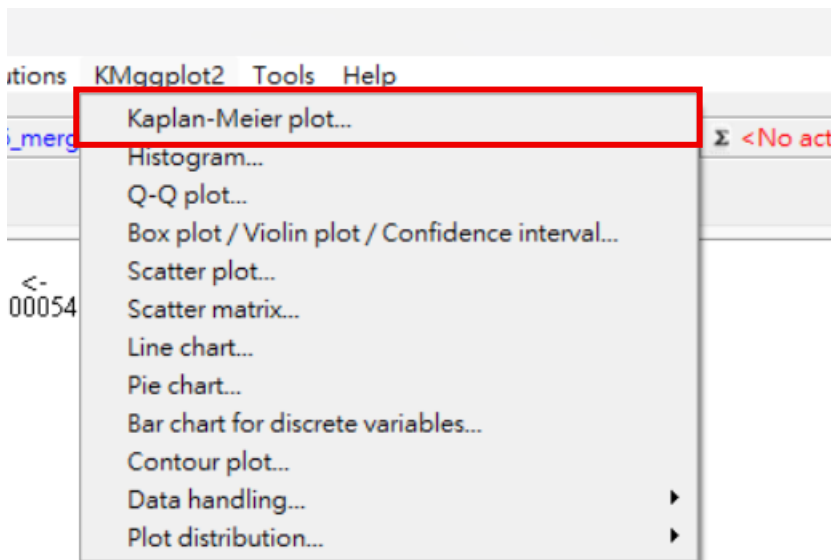
R Script R Markdown

Submit

載入檔案



選擇 Kaplan-Meier plot



Overall Survival(OS)：(要分析 OS_mean、OS_median、OS_quartile)

Kaplan-Meier plot

Time variable
Days.to.Last.Known.Disease.Status
Days.to.Recurrence
OS_Status
OS_Time
Pack.Years.Smoked
PFS_Status

Event variable (0=censor, 1=event)
OS_mean
OS_median
OS_quartile
OS_Status
OS_Time
Pack.Years.Smoked

Stratum variable
Last.Known.Disease.Status
Morphology
OS_mean
OS_median
OS_quartile
PFS_mean

Facet variable in rows
Age.Is.Obfuscated
AJCC.Pathologic.Stage
AJCC.Staging.System.Edition
Alcohol.History
Alcohol.Intensity
Case.ID

Facet variable in cols
Age.Is.Obfuscated
AJCC.Pathologic.Stage
AJCC.Staging.System.Edition
Alcohol.History
Alcohol.Intensity
Case.ID

Horizontal axis label
Time from entry

Vertical axis label
Proportion of survival

Legend label
<auto>

Title

No at risk
☒ None
☐ Inside
☐ Outside (facet ignored)

Tick count
4

Options
☒ Confidence interval (dotted lines)
☒ Confidence interval (band)
☐ Dot censored symbol
☒ P-value (log-rank test)
☐ Reference line at median survival

Font size
14

Font family
sans
serif
mono
Japan1
Japan1HeiMin

Colour pattern
Default
Hue
Grey
Set1
Blues

Graph options
☐ Save graph

Theme
theme_bw
theme_simple
theme_classic
theme_grey
theme_minimal

Help OK Cancel Preview

Stratum variable：

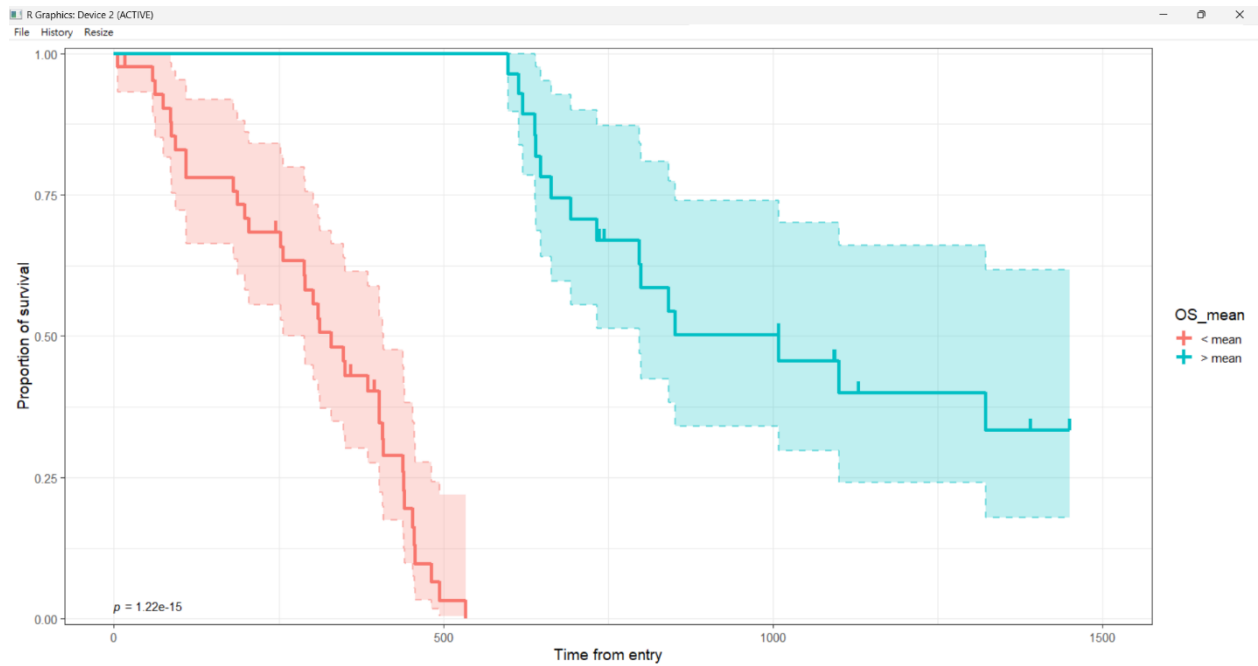
選擇要分析 OS 的類別

左方的 Options 可以依需求勾選

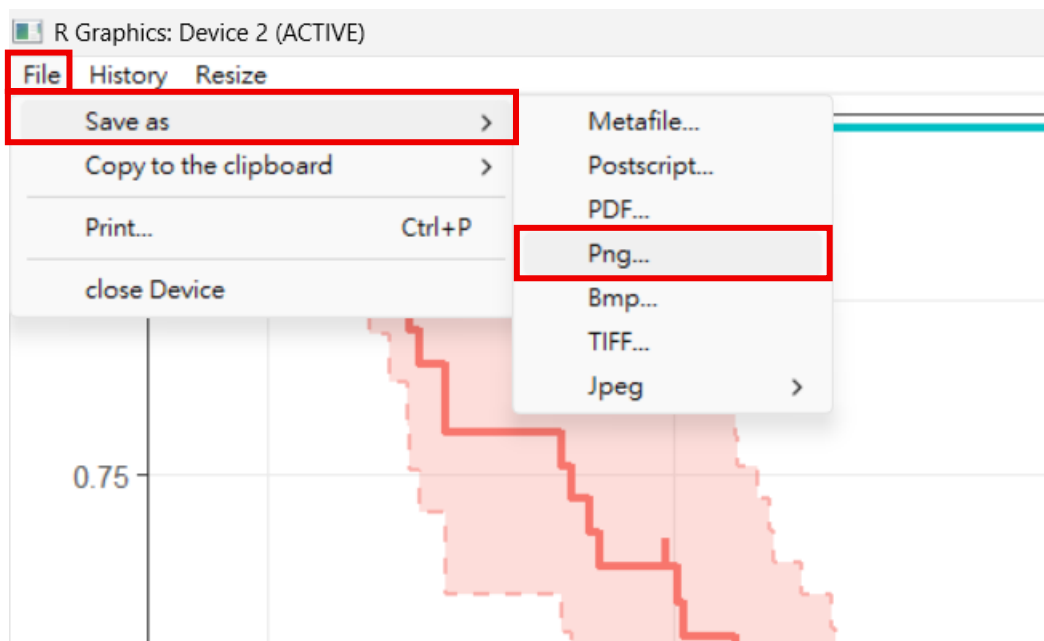
跑出結果，且此視窗會關掉，選項需要重選，適合跑單一分析時使用

跑出結果，但此視窗不會跳掉，選項不需要重選，適合跑多種分析時使用

跑出結果圖

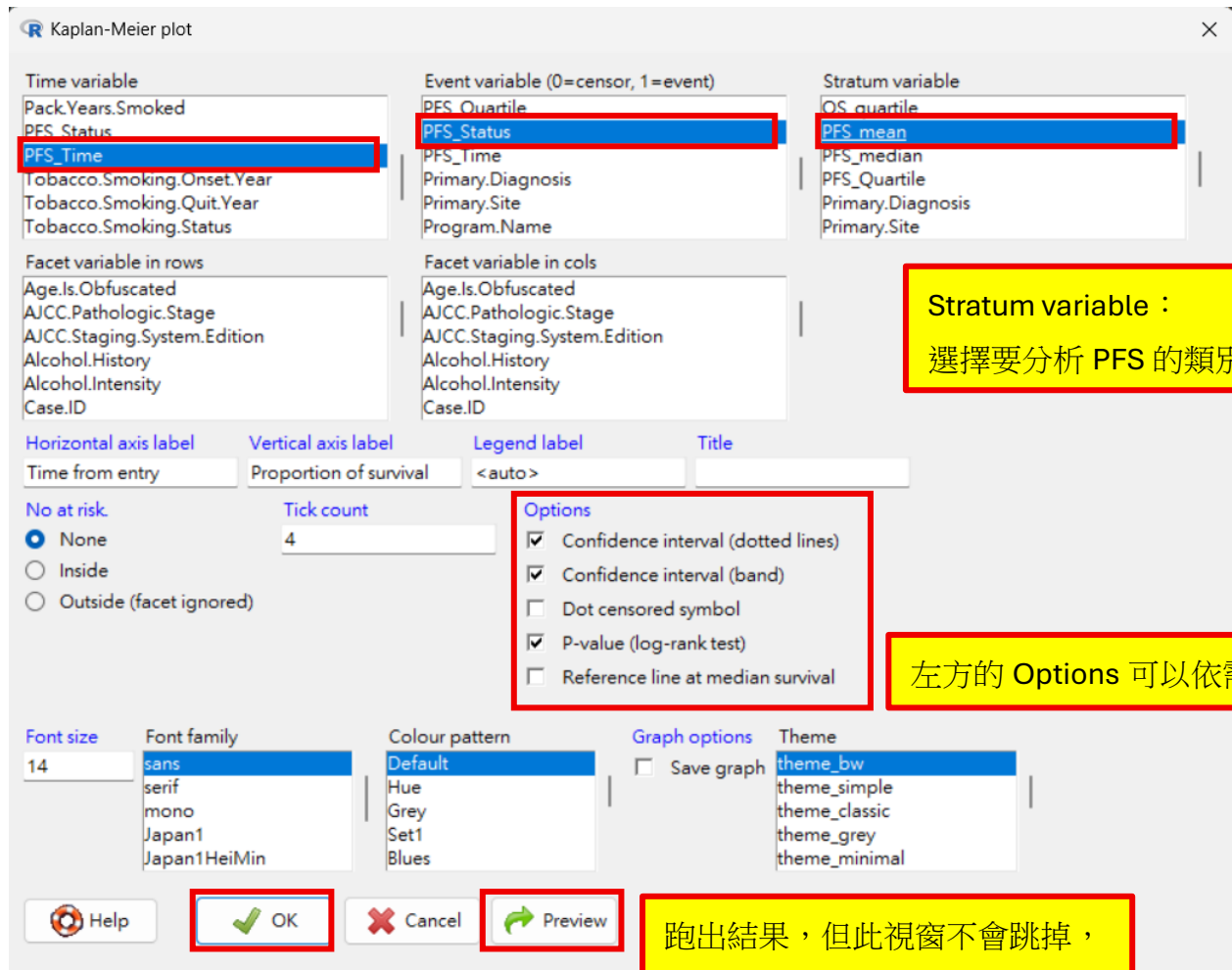


將結果圖存成 PNG 圖 (檔名：PDC000XXX_OS_XXX)



將 PNG 圖上傳到共享雲端中的 analysis result 中的對應位置

Progression Free Survival(PFS)：(要分析 PFS_mean、PFS_median、PFS_quartile)



The image shows the 'Kaplan-Meier plot' dialog box in R. The 'Time variable' list includes 'PFS_Time', which is highlighted with a red box. The 'Event variable (0=censor, 1=event)' list includes 'PFS_Status', also highlighted with a red box. The 'Stratum variable' list includes 'PFS_mean', highlighted with a red box. The 'Options' section is expanded and highlighted with a red box, showing checkboxes for 'Confidence interval (dotted lines)', 'Confidence interval (band)', 'Dot censored symbol', 'P-value (log-rank test)', and 'Reference line at median survival'. The 'OK' button is highlighted with a red box. The 'Preview' button is highlighted with a red box.

Time variable

Pack.Years.Smoked
PFS_Status
PFS_Time
Tobacco.Smoking.Onset.Year
Tobacco.Smoking.Quit.Year
Tobacco.Smoking.Status

Event variable (0=censor, 1=event)

PFS_Quartile
PFS_Status
PFS_Time
Primary.Diagnosis
Primary.Site
Program.Name

Stratum variable

OS_quartile
PFS_mean
PFS_median
PFS_Quartile
Primary.Diagnosis
Primary.Site

Facet variable in rows

Age.Is.Obfuscated
AJCC.Pathologic.Stage
AJCC.Staging.System.Edition
Alcohol.History
Alcohol.Intensity
Case.ID

Facet variable in cols

Age.Is.Obfuscated
AJCC.Pathologic.Stage
AJCC.Staging.System.Edition
Alcohol.History
Alcohol.Intensity
Case.ID

Horizontal axis label

Time from entry

Vertical axis label

Proportion of survival

Legend label

<auto>

Title

No at risk

☒ None
☐ Inside
☐ Outside (facet ignored)

Tick count

4

Options

☒ Confidence interval (dotted lines)
☒ Confidence interval (band)
☐ Dot censored symbol
☒ P-value (log-rank test)
☐ Reference line at median survival

Font size

14

Font family

sans
serif
mono
Japan1
Japan1HeiMin

Colour pattern

Default
Hue
Grey
Set1
Blues

Graph options

☐ Save graph

Theme

theme_bw
theme_simple
theme_classic
theme_grey
theme_minimal

Help

OK

Cancel

Preview

Stratum variable：

選擇要分析 PFS 的類別

左方的 Options 可以依需求勾選

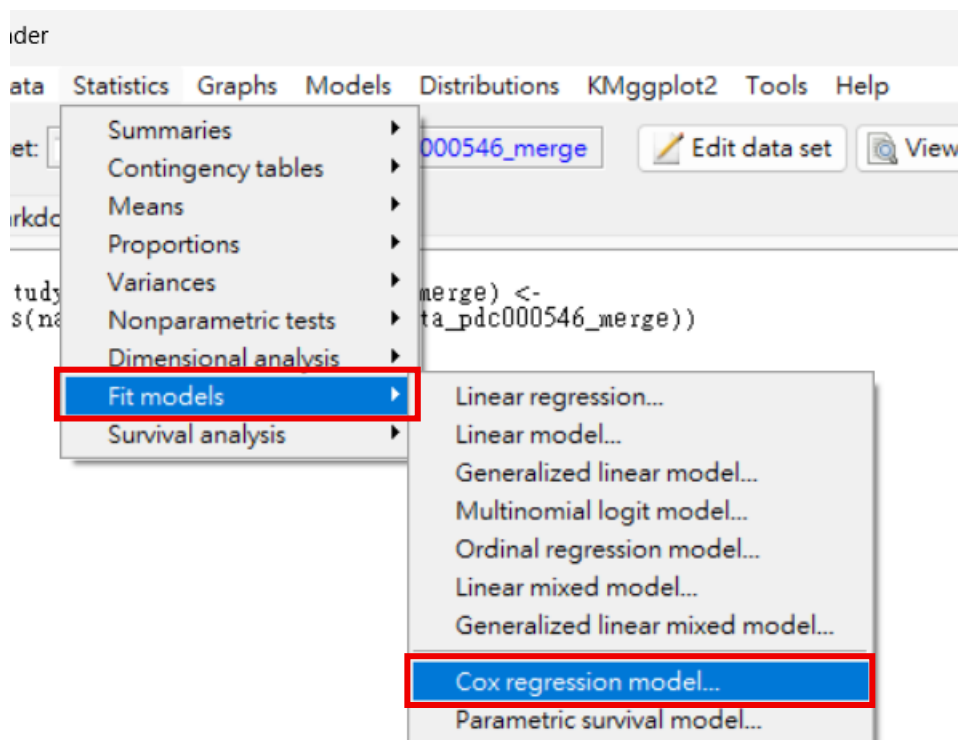
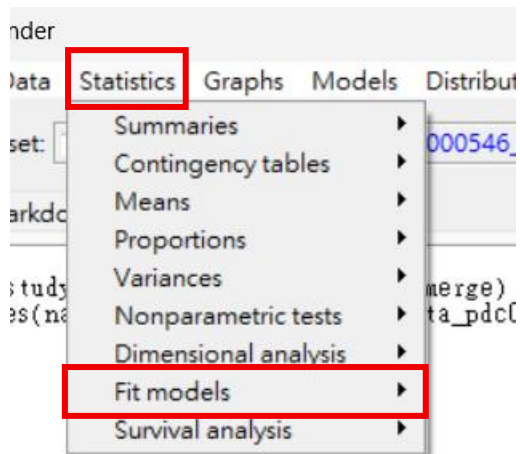
跑出結果，且此視窗會關掉，選項需要重選，適合跑單一分析時使用

跑出結果，但此視窗不會跳掉，選項不需要重選，適合跑多種分析時使用

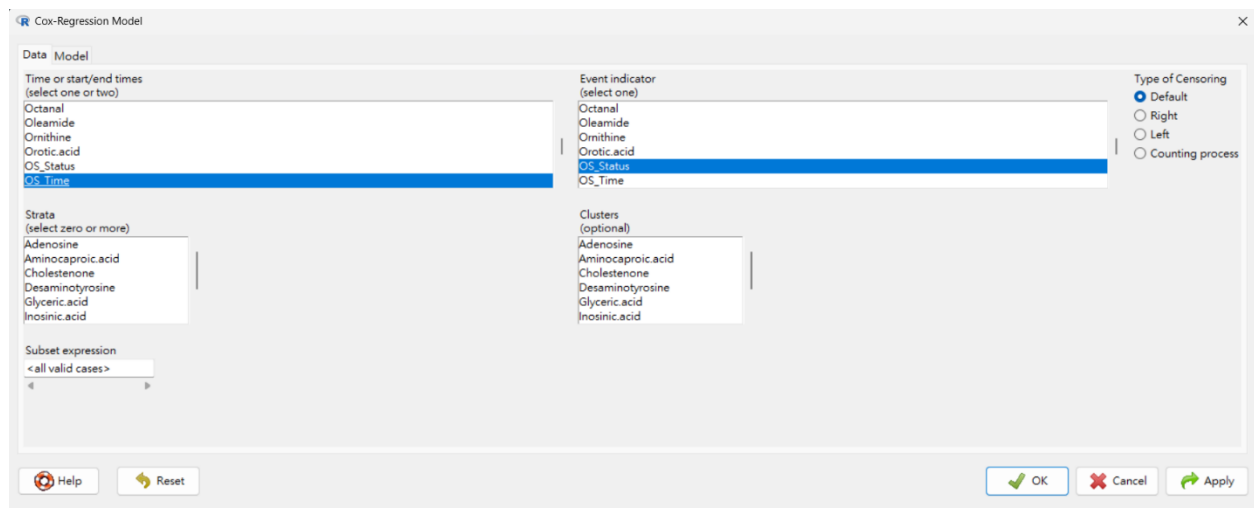
Cox regression (Hazard Ratio)

先將 Excel 檔 import 進 Rcmdr

再選擇 Cox regression model

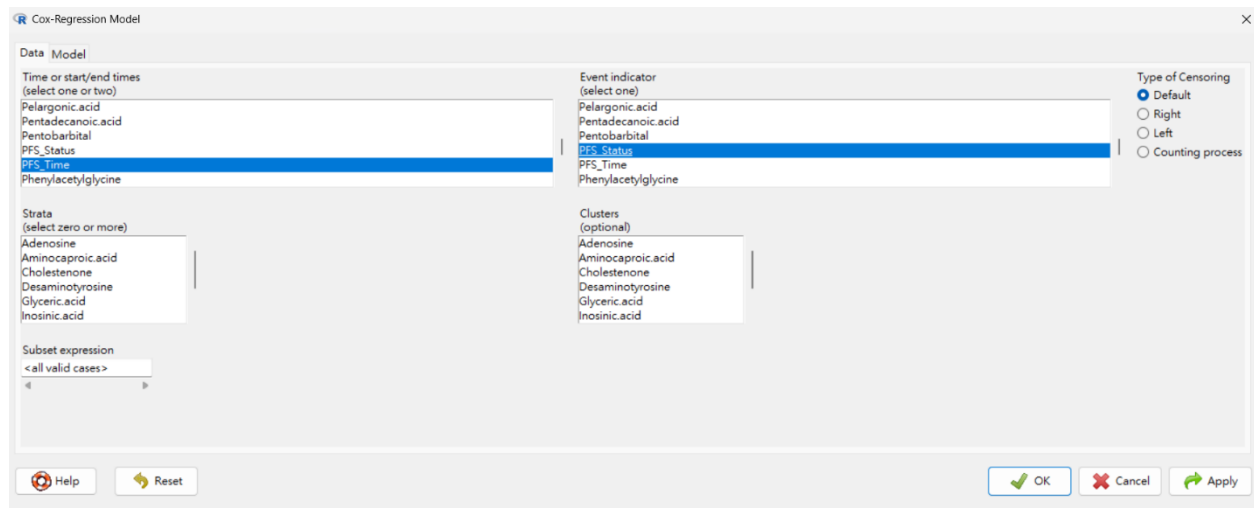


分析 metabolite 在 OS 的情況下的重要程度



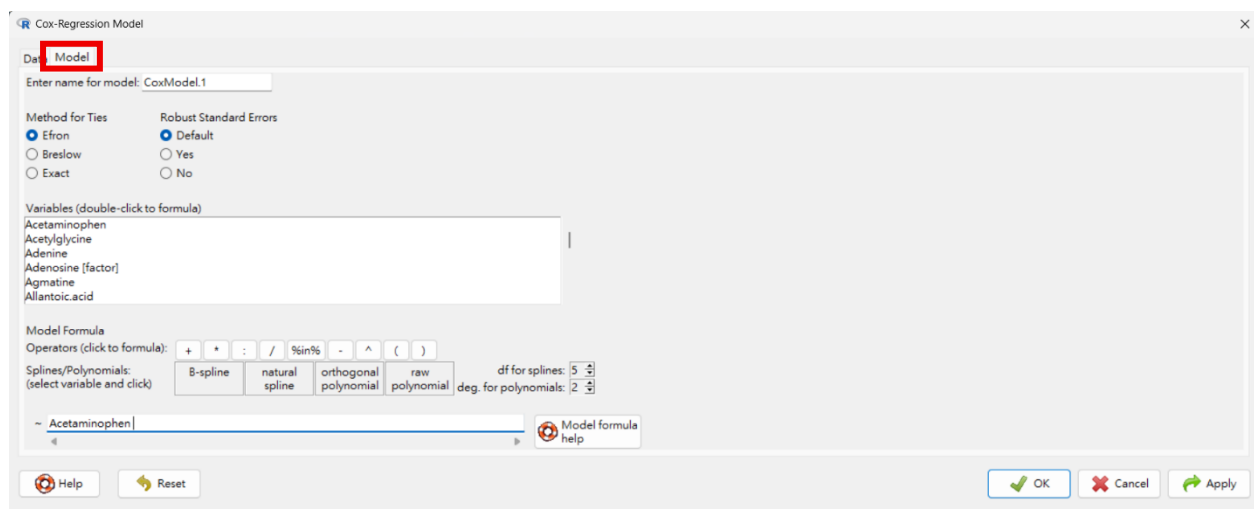
The 'Cox-Regression Model' dialog box is shown with the 'Data' tab selected. The 'Time or start/end times' list includes Octanal, Oleamide, Ornithine, Orotic.acid, OS_Status, and OS_Time, with OS_Time selected. The 'Event indicator' list includes Octanal, Oleamide, Ornithine, Orotic.acid, OS_Status, and OS_Time, with OS_Status selected. The 'Type of Censoring' is set to 'Default'. The 'Strata' list includes Adenosine, Aminocaproic.acid, Cholestenone, Desaminotyrosine, Glyceric.acid, and Inosinic.acid. The 'Subset expression' is set to '<all valid cases>'. The 'Clusters' list includes Adenosine, Aminocaproic.acid, Cholestenone, Desaminotyrosine, Glyceric.acid, and Inosinic.acid. The 'OK', 'Cancel', and 'Apply' buttons are at the bottom right.

分析 metabolite 在 PFS 情況下的重要程度



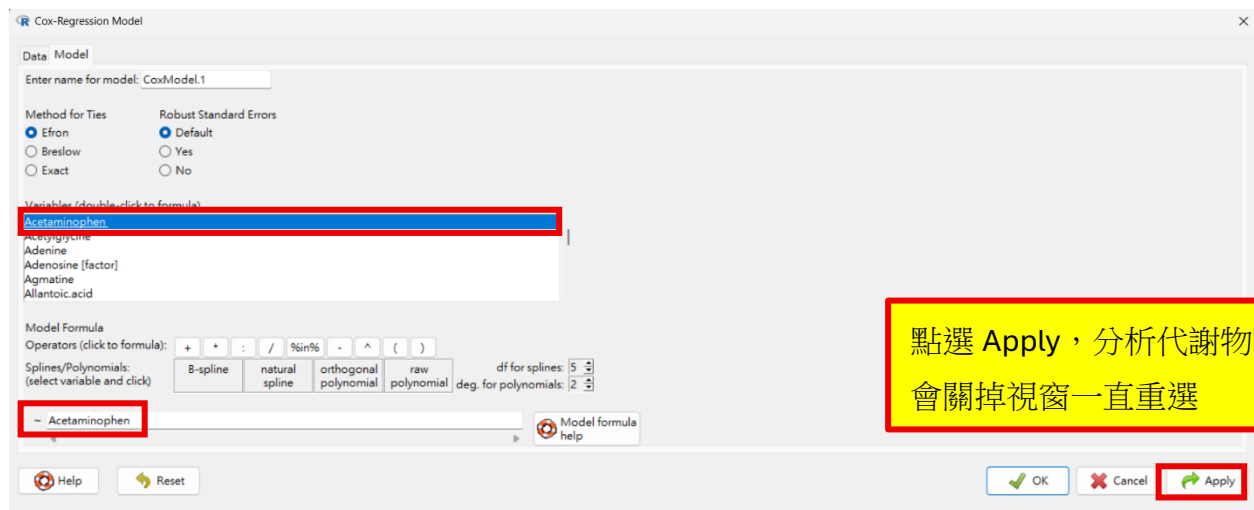
The 'Cox-Regression Model' dialog box is shown with the 'Data' tab selected. The 'Time or start/end times' list includes Pelargonic.acid, Pentadecanoic.acid, Pentobarbital, PFS_Status, PFS_Time, and Phenylacetylglutamine, with PFS_Time selected. The 'Event indicator' list includes Pelargonic.acid, Pentadecanoic.acid, Pentobarbital, PFS_Status, PFS_Time, and Phenylacetylglutamine, with PFS_Status selected. The 'Type of Censoring' is set to 'Default'. The 'Strata' list includes Adenosine, Aminocaproic.acid, Cholestenone, Desaminotyrosine, Glyceric.acid, and Inosinic.acid. The 'Subset expression' is set to '<all valid cases>'. The 'Clusters' list includes Adenosine, Aminocaproic.acid, Cholestenone, Desaminotyrosine, Glyceric.acid, and Inosinic.acid. The 'OK', 'Cancel', and 'Apply' buttons are at the bottom right.

選完在 OS 或 PFS 情況下後，點選 Model



The 'Cox-Regression Model' dialog box is shown with the 'Model' tab selected. The 'Enter name for model' field is set to 'CoxModel.1'. The 'Method for Ties' is set to 'Efron'. The 'Robust Standard Errors' are set to 'Default'. The 'Variables (double-click to formula)' list includes Acetaminophen, Acetylglutamine, Adenine, Adenosine [factor], Arginine, and Allantoinic.acid. The 'Model Formula' field is set to '~ Acetaminophen'. The 'Splines/Polynomials' section shows 'B-spline', 'natural spline', 'orthogonal polynomial', and 'raw polynomial' options. The 'df for splines' is set to 5 and 'deg. for polynomials' is set to 2. The 'OK', 'Cancel', and 'Apply' buttons are at the bottom right.

對要分析的 metabolite 點兩下，看到下方出現選擇的代謝物名稱



記得要分析其他代謝物時，要先”清空”再點兩下新增代謝物!!!

每次分析不會出現兩個或兩個以上代謝物!!!每次只會出現一個代謝物!!!

跳回去 R Studio 看左下方結果

結果 1. 當代謝物濃度直只有一個時，會出現 Error (不用紀錄結果)

```
RcmdrMsg+ D.Xylono.1.5.lactone, D.Xylulose, X1.3.Diaminopropane, X1.Kestose,
RcmdrMsg+ X1.Methyladenosine, MG.18.1.9Z..0.0.0.0., p.Hydroxymandelic.acid,
RcmdrMsg+ X5.Aminopentanoic.acid, N.Acetylserotonin,
RcmdrMsg+ Uridine.diphosphate.glucuronic.acid, cis.Sinapic.acid,
RcmdrMsg+ Dehydroabietic.acid, Erythronic.acid, Galacturonic.acid,
RcmdrMsg+ D.Glucosaminic.acid, Indoxyl.sulfate, Inosinic.acid,
RcmdrMsg+ X13.Methylmyristic.acid, beta.Lactose, D.Mannose, Nicotinic.acid,
RcmdrMsg+ Pentadecanoic.acid, Pimelic.acid, Glucaric.acid,
RcmdrMsg+ X2.Hydroxybenzaldehyde, D.Tagatose, Tartaric.acid, Terephthalic.acid,
RcmdrMsg+ Vanillic.acid, D.Xylose, D.Ribose,
RcmdrMsg+ X.2R.3R.4S.5S..2.3.4.5.Tetrahydroxyhexanal
RcmdrMsg: [11] ERROR: 3])
RcmdrMsg: [12] ERROR: object 'CoxModel.1' not found
RcmdrMsg: [13] ERROR: the model CoxModel.1 is no longer available
```

結果 2. 紀錄 n 數量、p value ($\Pr(>|z|)$)、hazard ratio ($\exp(\text{coef})$)

```
n= 2, number of events= 2
(75 observations deleted due to missingness)

      coef exp(coef) se(coef)      z Pr(>|z|)
Acetylglycine -0.000383 0.999617 0.726035 -0.001 1

      exp(coef) exp(-coef) lower .95 upper .95
Acetylglycine 0.9996      1      0.2409 4.148

Concordance= 1 (se = 0 )
Likelihood ratio test= 1.39 on 1 df,  p=0.2
Wald test              = 0 on 1 df,  p=1
Score (logrank) test = 1 on 1 df,  p=0.3
```

將結果紀錄在 dbMetabolite.xlsx 中的 Hazard Ratio 資料表中

dbMetabolite

Search for tools, help, and more (Alt + Q)

Home

Insert

Share

Page Layout

Formulas

Data

Review

View

Automate

Help

Draw

Comments

Catch up

Editing

Share

PMingLiu

12

B

General

€

¥

0.00

Σ

27

Q

Copilot

DMTDB ID	metabolite_name	病人預後 hazard ratio PDC000546-OS	PDC000546-OS n數量	PDC000546-OS P value	病人預後 hazard ratio PDC000546-PFS	PDC000546-PFS n數量	PDC000546-PFS P value	病人預後 hazard ratio PDC000552-OS	PDC000552-OS n數量	PDC000552-OS P value	病人預後 hazard ratio PDC000552-PFS
DMTDB000001	biphenyl										
DMTDB000002	nicotinamide										
DMTDB000003	Tranexamic Acid										
DMTDB000004	Quinic acid										
DMTDB000005	trans-trans-Muconic acid										
DMTDB000006	1,4-Dithiothreitol										
DMTDB000007	Parabanic Acid										
DMTDB000008	Nonadecanoic acid										
DMTDB000009	Octacosanoic acid										
DMTDB000010	Serotonin										
DMTDB000011	Palmitoleic acid										
DMTDB000012	Histidine										
DMTDB000013	4-Hydroxyproline										
DMTDB000014	N2-Acetylornithine										
DMTDB000015	Glycylproline										
DMTDB000016	Pipecolic acid										
DMTDB000017	Hydroquinone										
DMTDB000018	beta-Alanine										
DMTDB000019	Sumiki's acid										

main

browse

metabolite_info

Hazard Ratio

pubchem synonyms

HMDB synonyms

KEGG_PathwayName

+